Asthma Exacerbation, Shortness of Breath

The coefficient and standard error are based on an odds ratio of 1.08 (95% CI 1.00-1.17) for a 30 μg/m³ increase in 12-hour average $PM_{2.5}$ concentration (Ostro et al., 2001, Table 4, p.204).

Incidence Rate: daily shortness of breath rate per person (Ostro et al., 2001, p.202) = 0.074

Population: The study reported results for African-American population ages 8-13. For comparability to other studies, we apply the results to the African-American population of ages 6 to 18. We treat this as two groups based on the available information from American Lung Association (2010b, Table 9). Asthmatic African-American population ages 6 to 17 = 17.76% of African-American population ages 6 to 17 and asthmatic African-American population age 18 = 7.52% of African-American population age 18. The American Lung Association (2010b, Table 9) estimates asthma prevalence for children 5- 17 and adults 18-44 at 17.76% and 7.52% respectively (based on data from the 2008 National Health Interview Survey).

Asthma Exacerbation, Wheeze

The coefficient and standard error are based on an odds ratio of 1.06 (95% CI 1.01-1.11) for a 30 μg/m³ increase in 12-hour average $PM_{2.5}$ concentration (Ostro et al., 2001, Table 4, p.204).

Incidence Rate: daily wheeze rate per person (Ostro et al., 2001, p.202) = 0.173

Population: asthmatic African-American population ages 8 to 13 = 17.76% of African-American population ages 8 to 13. (Described above.)

G.1.6.3 Pope et al. (1991)

Using logistic regression, Pope et al. (1991) estimated the impact of PM_{10} on the incidence of a variety of minor symptoms in 55 subjects (34 "school-based" and 21 "patient-based") living in the Utah Valley from December 1989 through March 1990. The children in the Pope et al. study were asked to record respiratory symptoms in a daily diary. With this information, the daily occurrences of upper respiratory symptoms (URS) and lower respiratory symptoms (LRS) were related to daily PM_{10} concentrations. Pope et al. describe URS as consisting of one or more of the following symptoms: runny or stuffy nose; wet cough; and burning, aching, or red eyes. Levels of ozone, NO_2, and SO_2 were reported low during this period, and were not included in the analysis. The sample in this study is relatively small and is most representative of the asthmatic population, rather than the general population. The school-based subjects (ranging in age from 9 to 11) were chosen based on "a positive response to one or more of three questions: ever wheezed without a cold, wheezed for 3 days or more out of the week for a month or longer, and/or had a doctor say the 'child has asthma' (Pope et al., 1991, p. 669)." The patient-based subjects (ranging in age from 8 to 72) were receiving treatment for asthma and were referred by local physicians. Regression results for the school-based sample (Pope et al., 1991, Table 5) show PM_{10} significantly associated with both upper and lower respiratory symptoms. The patient-based sample did not find a significant PM_{10} effect. The results from the school-based sample are used here.

Upper Respiratory Symptoms

The coefficient and standard error for a one μg/m^3 change in PM_{10} is reported in Table 5.

Incidence Rate: daily upper respiratory symptom incidence rate per person = 0.3419 (Pope et al., 1991, Table 2)

Population: asthmatic population ages 9 to 11 = 10.70% of population ages 9 to 11. (The American Lung Association (2010b, Table 7) estimates asthma prevalence for children ages 5 to 17 at 10.70%, based on data from the 2008 National Health Interview Survey.)

G.2 Additional Ozone Health Impact Functions

Section G.2 summarizes the additional health impact functions for Ozone included in BenMAP-CE.

G.2.1 Short-term Mortality

Table G-11 summarizes the additional health impacts functions used to estimate the relationship between ozone and short-term mortality. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table G-11. Additional Health Impact Functions for Ozone and Mortality*

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Non-Accidental	Bell et al.	2004	95 US cities	0-99		D8HourMax	0.000261	0.000089	Log-linear	Warm season. 8-hour max from 24-hour mean
Non-Accidental	Bell et al.	2004	95 US cities	0-99		D24HourMean	0.000520	0.000128	Log-linear	All year
All Cause	Bell et al.	2005	US & non-US	0-99		D8HourMax	0.000795	0.000212	Log-linear	Warm season. 8-hour max from 24-hour mean
All Cause	Bell et al.	2005	US & non-US	0-99		D24HourMean	0.001500	0.000401	Log-linear	Warm season

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Cardio-pulmonary	Huang et al.	2005	19 US cities	0-99		D8HourMax	0.000813	0.000259	Log-linear	Warm season. 8-hour max from 24-hour mean.
Cardio-pulmonary	Huang et al.	2005	19 US cities	0-99		D24HourMean	0.001250	0.000398	Log-linear	Warm season
Non-Accidental	Ito and Thurston	1996	Chicago, IL	18-99	PM_{10}	D1HourMax	0.000634	0.000251	Log-linear	
Non-Accidental	Ito et al.	2005		0-99		D8HourMax	0.001173	0.000239	Log-linear	Warm season. 8-hour max from 24-hour mean.
Non-Accidental	Ito et al.	2005		0-99		D1HourMax	0.000400	0.000066	Log-linear	1-hour max
Non-Accidental	Ito et al.	2005		0-99		D24HourMean	0.001750	0.000357	Log-linear	Warm season. 24-hour mean
Non-Accidental	Ito et al.	2005		0-99		D8HourMax	0.000532	0.000088	Log-linear	8-hour max from 1-hour max
Long Term Mortality, Respiratory	Jerrett et al.	2009	86 urban areas	30-99	$PM_{2.5}$	Annual	0.003922	0.001325	Log-linear	
Long Term Mortality, Respiratory	Jerrett et al.	2009	Northeast	30-99		Annual	-0.001005	0.003853	Log-linear	
Long Term Mortality, Respiratory	Jerrett et al.	2009	Industrial Midwest	30-99		Annual	0.000000	0.004604	Log-linear	
Long Term Mortality, Respiratory	Jerrett et al.	2009	Southeast	30-99		Annual	0.011333	0.003193	Log-linear	
Long Term Mortality, Respiratory	Jerrett et al.	2009	Upper Midwest	30-99		Annual	0.013103	0.026212	Log-linear	
Long Term Mortality, Respiratory	Jerrett et al.	2009	Northwest	30-99		Annual	0.005827	0.003118	Log-linear	
Long Term Mortality, Respiratory	Jerrett et al.	2009	Southwest	30-99		Annual	0.019062	0.007583	Log-linear	
Long Term Mortality, Respiratory	Jerrett et al.	2009	Southern California	30-99		Annual	0.000995	0.002767	Log-linear	
Long Term Mortality, Respiratory	Jerrett et al.	2009	86 urban areas	30-99	$PM_{2.5}$	Annual	0.004471	0.001510	Log-linear	

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
All Cause	Levy et al.	2005	US and non-US	0-99		D8HourMax	0.001119	0.000179	Log-linear	Warm season. 8-hour max from 1-hour max.
All Cause	Levy et al.	2005	US and non-US	0-99		D1HourMax	0.000841	0.000134	Log-linear	Warm season.
Non-Accidental	Moolgavkar et al.	1995	Philadelphia, PA	0-99		D24HourMean	0.001398	0.000266	Log-linear	Warm season.
Non-Accidental	Moolgavkar et al.	1995	Philadelphia, PA	0-99	TSP, SO_2	D24HourMean	0.001389	0.000373	Log-linear	Warm season.
Non-Accidental	Moolgavkar et al.	1995	Philadelphia, PA	18-99	TSP, SO_2	D24HourMean	0.000611	0.000216	Log-linear	
Non-Accidental	Samet et al.	1997	Philadelphia, PA	18-99	CO, NO_2, SO_2, TSP	D24HourMean	0.000936	0.000312	Log-linear	
Non-Accidental	Schwartz	2005	14 U.S. cities	0-99		D8HourMax	0.000426	0.000150	Logistic	Warm season. 8-hour max from 1-hour max.
Non-Accidental	Schwartz	2005	14 U.S. cities	0-99		D1HourMax	0.000370	0.000130	Logistic	Warm season
Non-Accidental	Smith et al.	2009	98 U.S. cities	0-99		D8HourMax	0.000322	0.000084	Log-linear	Ozone season
Non-Accidental	Smith et al.	2009	98 U.S. cities	0-99	PM_{10}	D8HourMax	0.000258	0.000167	Log-linear	Ozone season
All-Cause	Zanobetti and Schwartz	2008	48 Cities	0-99		D8HourMax	0.00050	0.00012	Log-linear	0-3 day lag, June-August, 1989-2000

*Unless otherwise stated, mortality is short-term.

G.2.1.1 Bell et al. (2004)

Ozone has been associated with various adverse health effects, including increased rates of hospital admissions and exacerbation of respiratory illnesses. Although numerous time-series studies have estimated associations between day-to-day